

Arunabha Majumder

Mechanical Engineer | Sub-System Design & Ownership | Test Rigs | SolidWorks | Fabrication

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Hiring Team, Engineering
Rivan Industries
Bermondsey, London, United Kingdom

Application: Mechanical Engineer – Sub-System Owner

Dear Rivan Team,

I am applying for the Mechanical Engineer role. I hold a B.Tech in Mechanical Engineering and an M.Tech in Mechatronics, and I am finishing my PhD at Aalborg University, where for the past three years I have owned an electromechanical sub-system from concept to tested hardware. Owning a system end to end, and rethinking it from first principles, is exactly how I like to work.

In my PhD, I owned the concept, architecture and requirements of the actuation system for a wearable robot. I produced the SolidWorks models and drawings, built the parts hands-on through machining and additive manufacturing, and handled assembly, integration, commissioning and testing. Rethinking the actuation architecture, by pairing a parallel spring with a small 6 Nm motor, cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

I question every part's existence. On a 5-bar parallel robot, I removed the external force/torque sensor entirely by estimating force from the actuator itself, which reduced cost, mass, part count and integration complexity. I design test rigs to check hypotheses quickly before committing to fabrication, and I validated my analytical stiffness models against hardware to within 10% RMS error.

I am comfortable in the workshop. During my Master's at CSIR-CMERI, I fabricated pneumatic artificial muscles from raw materials and built the test setup with valves, pressure sensors and DAQ. Before that, I led the design and build of a Go-Kart from the ground up. I model in Python and MATLAB and use FEA to guide design decisions.

My thermofluids experience so far comes from pneumatic actuators rather than heat and mass transfer systems, and vendor management and procurement for a production line are new to me. These are the areas I want to grow into, and Rivan's mission to make synthetic fuel cheaper than fossil fuels is the kind of problem I want to work on.

I would welcome the opportunity to discuss how I can contribute to Rivan. References are available upon request.

Kind regards,
Arunabha Majumder